

LIE ALGEBRAS

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1. BASIC DEFINITIONS AND EXAMPLES

Definition 1.1. An *algebra* is a vector space over a field $\mathbb{F}$ endowed with a binary operation that is bilinear

$$\begin{aligned} a(\lambda b + \mu c) &= \lambda ab + \mu ac \\ (\lambda b + \mu c)a &= \lambda ba + \mu ca \end{aligned}$$

Definition 1.2. A *Lie algebra* is an algebra $\mathfrak{g}$ with a product $[\cdot, \cdot]$ we call *bracket* that satisfies

- (1) $[x, x] = 0 \quad \forall x \in \mathfrak{g},$
- (2) (Jacobi identity.)

Definition 1.3. A *simple Lie algebra* is a Lie algebra that has no nontrivial proper ideals.

As a rule of thumb I keep in mind the following silly computation:

$$\log \det A = \log \prod_i \lambda_i = \sum \log \lambda_i = \text{"trlog } A"$$

And recall that exponent map goes from $T_e G = \mathfrak{g} \rightarrow G$, so that logarithm would go from $G \rightarrow \mathfrak{g}$. This is why I remember that the condition on a classical Lie group of *having determinant 1* goes to *having vanishing trace* in the Lie algebra. (Because $\log 1 = 0$.)

Example 1.4. (1) The *special linear Lie algebra*

$$\mathfrak{sl}_n = \{\text{Mat}_n | \text{Tr}(A) = 0\}$$

which is just obvious from the slogan above.

(2) The *special orthogonal Lie algebra*

$$\mathfrak{so}_n = \{A \in \text{Mat}_n | A + A^T = 0\}$$

which is obvious from: $\text{SO}(n)$ = isometries, so $\langle v, v \rangle = \langle Av, Av \rangle = \langle v, A^T A v \rangle$, so $\text{SO}(n) = \{A \in \text{Mat}_n : A^{-1} = A^T\}$, and then $0 = \log 1 = \log(AA^T) = \log A + \log A^T$.

(3) The *symplectic Lie algebra*

$$\mathfrak{sp}_{2n} = \{A \in \text{Mat}_{2n} : \Omega A + A^T \Omega = 0\}$$

$$\text{where } \Omega = \begin{pmatrix} 0 & \text{Id}_n \\ -\text{Id}_n & 0 \end{pmatrix}.$$

Pending. Why this makes sense?

Example 1.5. If A is an associative Lie algebra, then A with the bracket $[a, b] = ab - ba$ is a Lie algebra, denoted by A_- . (It is an exercise to verify Jacobi's identity.) This gives $\mathfrak{gl}_V := \text{End}(V)_-$.

Definition 1.6. A *Lie subalgebra* is a vector subspace $\mathfrak{h} \subset \mathfrak{g}$ such that $[\mathfrak{h}, \mathfrak{h}] \subset \mathfrak{h}$.

Definition 1.7. A subspace $\mathfrak{h} \subset \mathfrak{g}$ is called an *ideal* if $[\mathfrak{h}, \mathfrak{g}] \subset \mathfrak{h}$.

Recall that a bilinear form $(\cdot, \cdot) : \mathfrak{g} \times \mathfrak{g} \rightarrow \mathbb{C}$ is *invariant* if $([a, b], c) = (a, [b, c])$ for all $a, b, c \in \mathfrak{g}$.

Definition 1.8. For any finite dimensional Lie algebra $\mathfrak{g}$ and $x \in \mathfrak{g}$ we write

$$\text{ad}_x = [x, -]$$

Exercise 1.9. Show that the map $\text{ad} : \mathfrak{g} \rightarrow \text{End}(\mathfrak{g})$, $x \mapsto \text{ad}_x$ is a representation, i.e. a morphism of Lie algebras.

Definition 1.10. The *Killing form* is

$$\kappa(x, y) = \text{Tr}_{\mathfrak{g}} \text{ad}_x \text{ad}_y$$

Exercise 1.11. Prove that an invariant bilinear form on a simple Lie algebra must in fact be symmetric.

Proof. (David.) It's enough to show that $\mathfrak{g}$ is *perfect*, i.e. that $[\mathfrak{g}, \mathfrak{g}] = \mathfrak{g}$. In this case, let $a, b \in \mathfrak{g}$ and suppose that $b = [x, y]$. Then

$$\begin{aligned} (a, b) &= (a, [x, y]) = (a, -[y, x]) = (-[a, y], x) = ([y, a], x) \\ &= (y, [a, x]) = (y, -[x, a]) = (-[y, x], a) = ([x, y], a) = (b, a) \end{aligned}$$

To confirm that $\mathfrak{g}$ is perfect just observe that $[\mathfrak{g}, \mathfrak{g}]$ is a nontrivial ideal of $\mathfrak{g}$. $\square$

Definition 1.12. A *semisimple Lie algebra* is a direct sum of simple ones

$$\mathfrak{g} = \mathfrak{g}_1 \oplus \dots \oplus \mathfrak{g}_s$$

Theorem 1.13 (Cartan). *The finite dimensional Lie algebra $\mathfrak{g}$ is semisimple if and only if its Killing form is nondegenerate.*

2. WEIGHT SPACES AND LIE'S THEOREM

Lie's theorem says that solvable Lie algebras over closed fields of characteristic not zero have weights.

Definition 2.1. Let $\mathfrak{h}$ be a Lie algebra, $\pi : \mathfrak{h} \rightarrow \mathfrak{gl}_V$ a representation of $\mathfrak{h}$ and $\lambda \in \mathfrak{h}^*$. The *weight space* of $\mathfrak{h}$ attached to λ is

$$V_{\lambda}^{\mathfrak{h}} := \{v \in V \mid \pi(h)v = \lambda(h)v, \forall h \in \mathfrak{h}\}.$$

If $V_{\lambda}^{\mathfrak{h}} \neq 0$, we say that λ is a *weight* for π .

My first impression on this concept is: a weight of a representation is a functional for which there is at least one vector (like an eigenvector) with the property that multiplying this vector times any other vector coming from the Lie algebra is a multiple of the vector (so the weight gives me kind of an eigenvalue).

Lemma 2.2 (Lie's Lemma). *Some condition on invariance used for...*

Theorem 2.3 (Lie's theorem). *Let $\mathfrak{g}$ be a solvable Lie algebra and π a representation of $\mathfrak{g}$ on a finite dimensional vector space $V \neq 0$, over an algebraically closed field $\mathbb{F}$ of characteristic 0. Then there exists a weight $\lambda \in \mathfrak{g}^*$ for π (that is, $V_\lambda^\mathfrak{g} \neq \{0\}$).*

Exercise 2.4. Show the following two corollaries of Lie's theorem:

- for all representations π of a solvable Lie algebra $\mathfrak{g}$ on a finite dimensional vector space V over an algebraically closed field $\mathbb{F}$, $\text{char}\mathbb{F} = 0$, there exists a basis for V for which the matrices of $\pi(\mathfrak{g})$ are upper triangular;
- a solvable subalgebra $\mathfrak{g} \subset \mathfrak{gl}_V$, (where V is finite-dimensional over an algebraically closed field $\mathbb{F}$, $\text{char}\mathbb{F} = 0$), is contained in the subalgebra of upper triangular matrices over $\mathbb{F}$ for some basis of V .

3. GENERALIZED EIGENSPACES AND GENERALIZED WEIGHT SPACES

Definition 3.1. A generalized eigenspace of $A \in \text{End}(V)$ with eigenvalue λ is

$$V_\lambda = \{v \in V : (A - \lambda \text{Id})^N = 0 \text{ for some positive integer } N\}.$$

It turns out that any linear operator on an algebraically closed field gives a decomposition into generalized eigenspaces via Jordan canonical form:

Proposition 3.2. *Let A be a linear operator on a finite-dimensional vector space V over an algebraically closed field $\mathbb{F}$, and let $\lambda_1, \dots, \lambda_s$ be all eigenvalues of A , and $n_1, \dots, n_s$ their multiplicities. Then one has the generalized eigenspace decomposition:*

$$V = \bigoplus_{i=1}^s V_{\lambda_i}, \quad \dim V_{\lambda_i} = n_i.$$

In particular, for a Lie algebra $\mathfrak{g}$ with a representation (i.e. Lie algebra morphism) $\pi : \mathfrak{g} \rightarrow \text{End}(V)$ we have

$$V = \bigoplus_{\lambda \in \mathbb{F}} V_\lambda^a, \quad V_\lambda^a = \{v \in V : (\pi(a)v - \lambda \text{Id})^N v = 0 \text{ for some } N \in \mathbb{N}\}$$

And even more particularly, for the adjoint representation we have

$$V = \bigoplus_{\lambda \in \mathbb{F}} \mathfrak{g}_\lambda^a, \quad \mathfrak{g}_\lambda^a = \{b \in \mathfrak{g} : ([a, \cdot] - \lambda \text{Id})^N b = 0, \text{ for some } N \in \mathbb{N}\}$$

And a little more generally we have

Definition 3.3. Let $\mathfrak{h}$ be a Lie algebra with a representation π on a vector space V , and $\lambda \in \mathfrak{h}^*$. A generalized weight space of $\mathfrak{h}$ in V attached to λ is

$$V_\lambda^\mathfrak{h} = \left\{ v \in V : (\pi(a) - \lambda(a)\text{Id})^N v = 0, \begin{array}{l} \text{for some } N \in \mathbb{N} \\ \text{depending on } a \in \mathfrak{h} \\ \text{for all } a \in \mathfrak{h} \end{array} \right\}$$

Under the right conditions, a nilpotent subalgebra $\mathfrak{h} \subset \mathfrak{g}$ permits decomposing V as a direct sum of generalized weight spaces of $\mathfrak{h}$. Namely,

Theorem 3.4. *Let $\mathfrak{g}$ be a finite-dimensional Lie algebra and π its representation on a finite-dimensional vector space V , over an algebraically closed field $\mathbb{F}$ of characteristic 0. Let $\mathfrak{h}$ be a nilpotent subalgebra of $\mathfrak{g}$. Then the following equalities hold:*

$$V = \bigoplus_{\lambda \in \mathfrak{h}^*} V_{\lambda}^{\mathfrak{h}}$$

$$\pi(\mathfrak{g}_{\alpha}^{\mathfrak{h}}) V_{\lambda}^{\mathfrak{h}} \subseteq V_{\lambda+\alpha}^{\mathfrak{h}}$$

Which in the case of the adjoint representation, looks like:

$$\mathfrak{g} = \bigoplus_{\alpha \in \mathfrak{h}^*} \mathfrak{g}_{\alpha}^{\mathfrak{h}}$$

$$[\mathfrak{g}_{\alpha}^{\mathfrak{h}}, \mathfrak{g}_{\beta}^{\mathfrak{h}}] \subseteq \mathfrak{g}_{\alpha+\beta}^{\mathfrak{h}}$$

4. SEMISIMPLE LIE ALGEBRAS

Definition 4.1. A *radical* $R(\mathfrak{g})$ of a finite-dimensional Lie algebra $\mathfrak{g}$ is a solvable ideal of $\mathfrak{g}$ of maximal possible dimension.

Proposition 4.2. *The radical ideal of $\mathfrak{g}$ contains any solvable ideal of $\mathfrak{g}$ and is unique.*

If $\mathfrak{g}$ is a finite dimensional solvable Lie algebra, then $R(\mathfrak{g}) = \mathfrak{g}$. The opposite case is when $R(\mathfrak{g}) = 0$.

Definition 4.3. A finite-dimensional Lie algebra $\mathfrak{g}$ is called *semisimple* if $R(\mathfrak{g}) = 0$.

In [Kac10, Lecture 11] we have the tools needed for a characterization of semisimple Lie algebras in terms of the Killing form:

Theorem 4.4 (Cartan). *Let $\mathfrak{g}$ be a finite-dimensional Lie algebra over a field of characteristic 0. Then the Killing form on $\mathfrak{g}$ is non-degenerate if and only if $\mathfrak{g}$ is semisimple. Moreover, if $\mathfrak{g}$ is semisimple and $\mathfrak{a} \subset \mathfrak{g}$ is an ideal, then the restriction of the Killing form to $\mathfrak{a}$, $K|_{\mathfrak{a} \times \mathfrak{a}}$, is also nondegenerate and coincides with the Killing form of $\mathfrak{a}$.*

REFERENCES

- [Kac10] Victor Kac, *Lecture notes of 18.745 – introduction to lie algebras (fall 2010)*, <https://math.mit.edu/classes/18.745/classnotes.html>, 2010, Lecture notes, MIT.